Tanta University
Faculty of Computers and Informatics
Second Level – Second Term (2025 -2026)

DIGITAL SIGNALS PROCESSING

Dr. Marwa H. Salama

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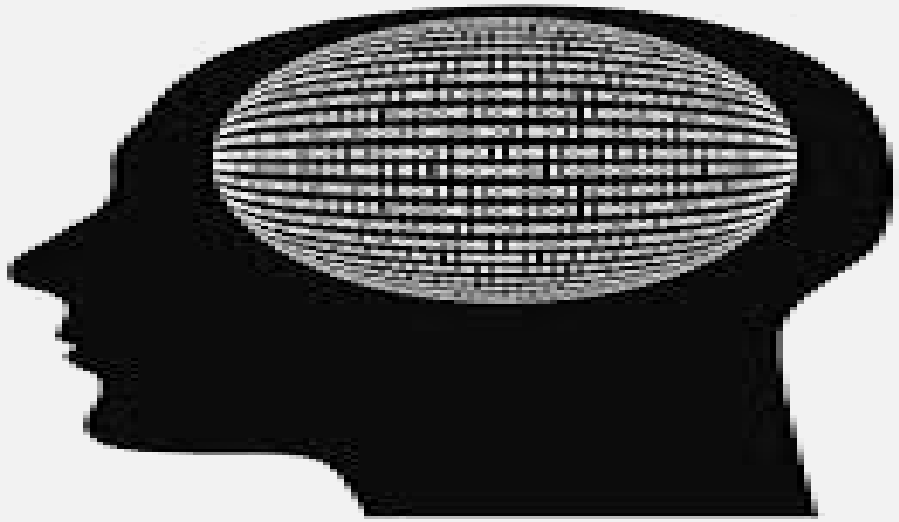
EXPECTED PERCEPTIONS



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COURSE MOTIVATION

This course provides students with fundamental concepts and practical skills essential for understanding and analyzing digital signals. It supports advanced courses in communications, image processing, and artificial intelligence, and equips students with analytical and problem-solving skills required in modern technological applications.

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TEACHING AND LEARNING METHODS

- **A mixture of will teach the course:**

➢ Theoretical Lectures.

➢ Supervised Tutorial Session

➢ Self-Study Group-Work.

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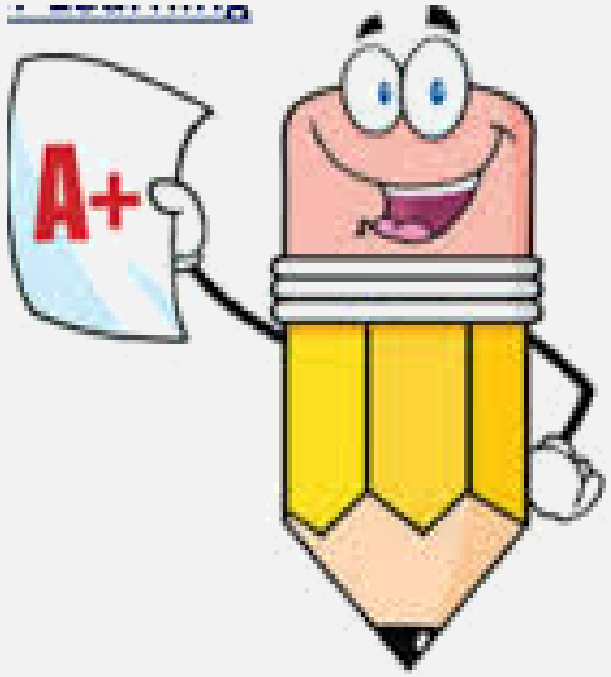
ASSESSMENT AND GRADING SYSTEM

- Mid Term Exam 20 Marks
- Online Quizzes/ Sheets 20 Marks
- Final Exam 60 Marks

- **Total 100 Marks**



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COURSE CONTENTS

1. Introduction
2. Discrete Time Signals and Systems

- Continuous vs. discrete time signals
- Standard Discrete Time
- Signals Discrete Time Systems (Operations and Classification)
- Linear Convolution

3. Z-Transform
4. Fourier Transform
5. Fast Fourier Transform
6. Digital Filter Structures
7. Filter Design

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Chapter 1: Introduction

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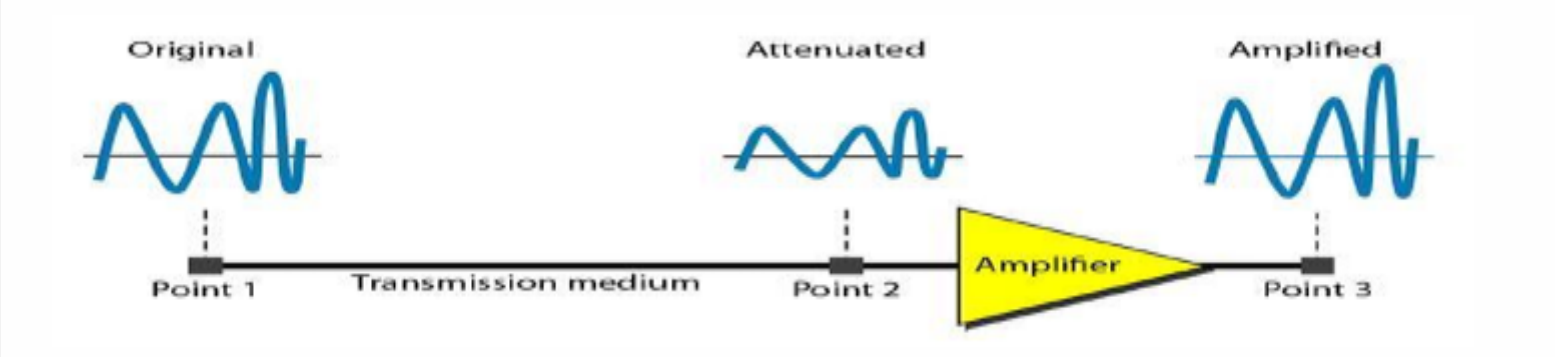
DIGITAL SIGNAL PROCESSING (DSP)

- DSP is afield of engineering that concerns with digital signals and the processes performed on these signals using digital systems.
- It performs operations on digital data like: amplification, attenuation, filtering, spectral analysis, feature extraction etc..
- The computer performs any operation digitally, hence it is considered a digital signal processor or digital system.



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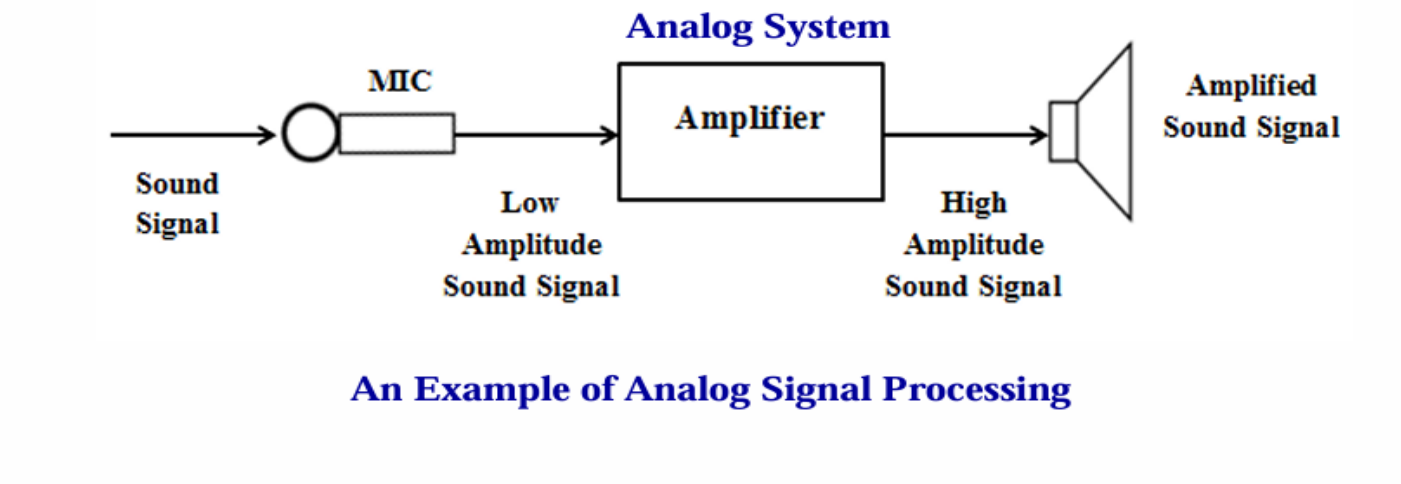
TYPES OF SIGNAL PROCESSING

(1) Analog Signal Processing:

- In such type of signal processing the input signal, the output signal and the system are all analog in nature.



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TYPES OF SIGNAL PROCESSING

(2) Digital Signal Processing

- In such type of signal processing the input signal, the output signal and the system are all digital in nature.
- For the following example ,the input signal is a noisy fingerprint image, and the output is filtered or noiseless fingerprint image. The filtering operation is done by the digital signal processor or the computer.

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BASIC ELEMENTS IN DIGITAL SIGNAL PROCESSING

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BASIC ELEMENTS IN DSP SYSTEMS

- A/D (Analog to digital converter) : Converts analog input to digital input.
- Digital Signal Processor or Digital System : it performs operations on digital data like: amplification, attenuation, filtering, spectral analysis, feature extraction etc. , the output signal of digital signal processor is called processed signal.
- D/A (digital to Analog converter) : Some of the processed Signals are required back in their analog form. For example sound, image and video signals are required in analog form. The D/A converter converts digital output of DSP processor to its analog equivalent.

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ADVANTAGES OF DSP SYSTEMS OVER ANALOG SIGNAL PROCESSING

- Flexibility: DSP systems are flexible. The system can be reconfigured for some other operation by simply changing the software program. For example, the high pass digital filter can be changed to low pass digital filter by simply changing the software.
- Accuracy.
- Easy storage.
- Cost: When there is a large complexity in the application, then DSP systems are cheaper compared to analog systems.
- Small in size and more reliable.
- Easily upgradable (since they are software controlled)

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DISADVANTAGES OF DSP SYSTEMS

1) When the analog signals have wide bandwidth, then high speed A/D converters are required. Such high speeds of A/D conversion are difficult to achieve.

2) The DSP systems are expensive for small applications. Hence the selection is done on the basis of cost complexity and performance.

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DSP APPLICATIONS

- DSP for Voice and Speech: Speech recognition, voice mail, speech enhancement, speaker verification, text to speech etc.
- DSP for Graphics and Imaging: animation, image and pattern recognition, image enhancement, robot vision, satellite imaging for multipurpose applications etc.
- DSP for Biomedical Engineering: ECG analysis, EEG brain mappers, Ultrasound equipment, patient monitoring systems, diagnostic tools etc.
- DSP for Consumer Applications: Digital audio/video/Television/Music systems, Toys etc.
- DSP for Control Applications: robot control, servo control, motor control etc.

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QUESTIONS

1) Explain the main difference between analog signal processing and digital signal processing.
2) Explain the basic elements of DSP systems.
3) What is meant by flexibility property of DSP system?
4) What are the disadvantages of DSP Systems?
5) List some examples of DSP applications in Biomedical Engineering field.

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